

A Simple Alternating Framework for ReLU-based Nonlinear Tensor Decomposition

Qingsong Wang, Junjie Yuan

Abstract—We study a nonlinear low-rank tensor decomposition model in which the observed nonnegative tensor $\mathcal{M}$ is generated by rectifying a latent low-rank tensor $\mathcal{X}$ element-wise via the ReLU function, i. e., $\mathcal{M} \approx \max(0, \mathcal{X})$. This paradigm is motivated by data that are sparse and nonnegative in the ambient space but admit a much more compact representation before rectification. To solve the resulting constrained nonconvex model, we introduce a simple alternating minimization framework, which denoted by NTD-tSVD. We also briefly analyze the computational complexity and convergence of the proposed algorithm. Extensive experiments on synthetic and real datasets demonstrate consistent improvements over direct linear low-rank approximations, yielding markedly lower reconstruction discrepancy in the rectified domain, especially when the observations are highly sparse. Moreover, the code is available at <https://github.com/nothing2wang/NTD-tSVD>.

Index Terms—Alternating minimization, Low-rank, Nonlinear tensor decomposition.

I. INTRODUCTION

TENSORS—as higher-order generalizations of vectors and matrices [1], [2]—have become indispensable for modeling multi-way data across a wide spectrum of applications, including machine learning [3], [4], image and signal processing [5], [6], [7]. In imaging in particular, color images and grayscale videos are naturally represented as third-order tensors, while color videos give rise to fourth-order tensors. Real-world tensor data frequently display low-rank structure because the intrinsic dimensionality of the underlying factors is much smaller than the ambient size.

Given a 3rd-order tensor $\mathcal{M} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, a common assumption is that it admits a good low-rank approximation, i.e.,

$$\mathcal{M} \approx \mathcal{X}, \quad \mathcal{X} \text{ is low-rank}, \quad (1)$$

which underlies many tensor decomposition and completion methods. Unlike the matrix case, tensor rank is model-dependent, with popular choices including CANDECOMP/PARAFAC (CP) decomposition [8], Tucker [9], tensor train (TT) [10], [11] and more general tensor network constructions [12]. Beyond these classical models, there also exist neural network-based tensor completion methods [13]

This research is supported by the National Natural Science Foundation of China (No. 12401415), the 111 Project (No. D23017), the Natural Science Foundation of Hunan Province (No. 2025JJ60009). (Corresponding author: Qingsong Wang)

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and robust low-rank matrix recovery approaches based on (nonconvex) Schatten-norm minimization [7], [14], [15] that explicitly handle outliers or gross corruptions.

The most common approach [1], [15] is to directly approximate the data with a low rank. However, this is not always appropriate. For example, for some sparse non-negative data¹, which may have a relatively high rank, directly approximating with a low rank can cause a large error, or a relatively large rank R may be required for approximation. See a real examples in Figure 1.

To address this, we advocate a nonlinear tensor decomposition (NTD) paradigm tailored to nonnegative sparse tensors. Specifically, for a nonnegative sparse tensor $\mathcal{M} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, we seek a real-valued low-rank tensor $\mathcal{X}$ of equal or lower rank such that

$$\mathcal{M} \approx f(\mathcal{X}), \quad \mathcal{X} \text{ is low-rank},$$

where function f acts element-wise on $\mathcal{X}$. Of particular interest is the rectified linear unit (ReLU)², $f(x) = \max(0, x)$, which is ubiquitous in modern deep networks [16]. This yields the specialized model

$$\mathcal{M} \approx \max(0, \mathcal{X}), \quad \mathcal{X} \text{ is low-rank}. \quad (2)$$

The formulation (2) confers a key advantage on sparse $\mathcal{M}$: entries that are zero in $\mathcal{M}$ correspond to nonpositive values in $\mathcal{X}$, thereby expanding the feasible set in the latent space and providing additional flexibility for uncovering compact low-rank structure.

We illustrate the motivation with a real data experiment (Figure 1). On the sparse, nonnegative *MNIST* tensor ($784 \times 5000 \times 10$), a rank-70 approximation via the classical low-rank model (1) using t-SVD [17], [18] achieves PSNR= 21.48, whereas our ReLU-NTD instantiation of (2) (Algorithm 1) attains PSNR= 27.58, recovering substantially clearer low-rank structure.

To quantify the discrepancy between $\mathcal{M}$ and $\max(0, \mathcal{X})$, we consider the following nonconvex program:

$$\min_{\mathcal{X}} \frac{1}{2} \|\mathcal{M} - \max(0, \mathcal{X})\|_F^2, \quad \text{s.t. rank}(\mathcal{X}) = R, \quad (3)$$

where R is typically much smaller than the ambient dimensions. We refer to (3) as ReLU-NTD.

The matrix case has recently drawn attention. Saul [19], [20] proposed an analogous ReLU-based nonlinear matrix

¹<http://frostd.io/tensors/>

²We hypothesize the existence of an underlying tensor $\mathcal{X}$ satisfying: (i) $\mathcal{X}$ is low rank; (ii) $\mathcal{X}_{i_1 \dots i_N} = \mathcal{M}_{i_1 \dots i_N}$ whenever $\mathcal{M}_{i_1 \dots i_N} > 0$; (iii) $\mathcal{X}_{i_1 \dots i_N} \leq 0$ whenever $\mathcal{M}_{i_1 \dots i_N} = 0$. Among simple element-wise maps, ReLU uniquely matches these requirements.

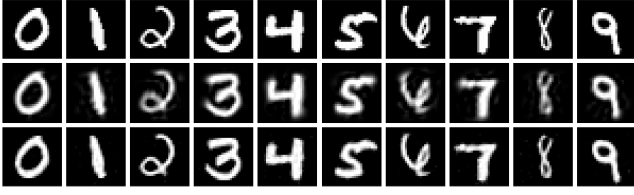


Fig. 1: Low-rank approximation for the *MNIST* dataset ($784 \times 5000 \times 10$) with $\text{rank}(\mathcal{X}) = 70$. **Top:** original data. **Middle:** model (1) via t-SVD [17], [18], PSNR= 21.48. **Bottom:** model (2) via Algorithm 1, PSNR= 27.58.

decomposition (ReLU–NMD) together with dedicated solvers. Subsequent works [21], [22], [23] improved computational efficiency and demonstrated utility in, e.g., basis compression [21] and clustering [23]. Nevertheless, these developments primarily target matrices, leaving higher-order tensors largely open—bridging this gap is a central aim of our study.

Directly minimizing (3) is challenging due to the composite nonlinearity embedded in a nonconvex low-rank model. To decouple the ReLU map from the low-rank constraint, we introduce an auxiliary tensor $\mathcal{W}$ and arrive at the following equivalent constrained problem:

$$\begin{aligned} \min_{\mathcal{W}, \mathcal{X}} \quad & \frac{1}{2} \|\mathcal{W} - \mathcal{X}\|_F^2, \\ \text{s.t.} \quad & \text{rank}(\mathcal{X}) = R, \max(0, \mathcal{W}) = \mathcal{M}. \end{aligned} \quad (4)$$

This splitting isolates the element-wise nonlinearity in the feasibility set while retaining a simple quadratic objective, paving the way for efficient alternating and projection-based algorithms developed in the sequel.

Contribution. In this paper, we establish a robust framework in (4), and introduce an alternating algorithm (Algorithm 1) tailored to address it. Specifically:

- **Model & Algorithm.** We formalize a ReLU-based nonlinear low-rank tensor decomposition in which the observation $\mathcal{M}$ is the element-wise rectification of a latent rank- R tensor $\mathcal{X}$, i.e., $\mathcal{M} \approx \max(0, \mathcal{X})$, and design a parameter-light alternating minimization scheme via projection–tSVD splitting. The $\mathcal{W}$ -step admits a closed-form projection onto the ReLU-feasible set, while the $\mathcal{X}$ -step is a rank- R approximation computed by truncated tSVD in the Fourier domain, yielding per-iteration cost dominated by FFTs and slice-wise SVDs.
- **Experiments.** On synthetic tensors (with controllable sparsity) and real datasets, the proposed method consistently improves rectified-domain reconstruction over direct linear low-rank baselines under the same target rank, particularly in highly sparse regimes.

The subsequent sections of this paper are structured as follows: Section II provides an in-depth exposition of the proposed algorithm (Algorithm 1). Section III utilizes synthetic and real datasets to demonstrate the stability of model (4) and the efficacy of the proposed algorithm, respectively. Finally, we draw conclusions in Section IV.

Notations. We use script characters (e.g. $\mathcal{X}, \mathcal{M}$) to denote tensors with at least three modes, capital letters (e.g. A, B) for

matrices. For $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, $\|\mathcal{X}\|_F = \sqrt{\sum_{i,j,k} \mathcal{X}_{ijk}^2}$. “fft” and “ifft” denote the Fourier transform and inverse Fourier transform, respectively. They satisfy $\text{ifft}(\text{fft}(\mathcal{X}, [], 3), [], 3) = \mathcal{X}$. Given $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and $\mathcal{B} \in \mathbb{R}^{I_2 \times I_4 \times I_3}$, the t-product $\mathcal{C} = \mathcal{A} * \mathcal{B} \in \mathbb{R}^{I_1 \times I_4 \times I_3}$, which corresponds to matrix-by-matrix multiplication of the frontal slices in the Fourier domain. $\mathcal{X}^T$ is the transpose of $\mathcal{X}$. See Figure 2 for an intuitive illustration of the tSVD factorization [18].

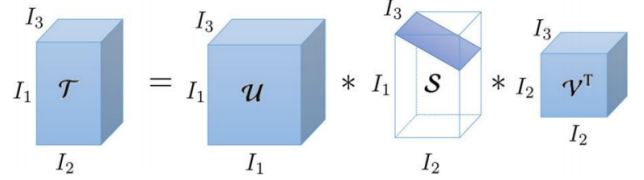


Fig. 2: An illustration of the tSVD of an $I_1 \times I_2 \times I_3$ tensor [24].

II. ALGORITHM

In this section, We solve problem (4) by exact two-block alternating minimization with variable splitting between the elementwise ReLU feasibility and the low-rank (t-rank) constraint

$$\begin{cases} \mathcal{W}^{k+1} = \arg \min_{\max(0, \mathcal{W}) = \mathcal{M}} \frac{1}{2} \|\mathcal{W} - \mathcal{X}^k\|^2, \\ \mathcal{X}^{k+1} = \arg \min_{\text{rank}(\mathcal{X}) = R} \frac{1}{2} \|\mathcal{W}^{k+1} - \mathcal{X}\|^2. \end{cases}$$

Let $\Omega_+ := \{(i, j, k) : \mathcal{M}_{ijk} > 0\}$ and $\Omega_0 := \{(i, j, k) : \mathcal{M}_{ijk} = 0\}$. The $\mathcal{W}$ -subproblem is a closed-form projection onto the ReLU-feasible set

$$\mathcal{W}_{i_1 i_2 i_3}^{k+1} = \begin{cases} \min(0, \mathcal{X}_{i_1 i_2 i_3}^k), & \text{if } (i_1, i_2, i_3) \in \Omega_0, \\ \mathcal{M}_{i_1 i_2 i_3}, & \text{if } (i_1, i_2, i_3) \in \Omega_+. \end{cases} \quad (5)$$

Note that the elements in $\mathcal{W}$ and $\mathcal{X}$ are not required to be nonnegative. The $\mathcal{X}$ -subproblem is the best rank- R t-approximation of $\mathcal{W}^{k+1}$, i.e., $\mathcal{X}^{k+1} = \Pi_{\mathcal{B}_R}(\mathcal{W}^{k+1})$ where $\mathcal{B}_R := \{\mathcal{X} : \text{rank}_t(\mathcal{X}) \leq R\}$. It is computed by truncated t-SVD: apply FFT along mode-3, perform rank- R truncated SVD on each frontal slice in the Fourier domain, and invert the FFT. Now we give the NTD-tSVD algorithm (Algorithm 1) framework as follows.

Computational complexity: For an $I_1 \times I_2 \times I_3$ tensor and target rank R , one iteration has:

- **$\mathcal{W}$ -step:** element-wise projection (5), $\mathcal{O}(I_1 I_2 I_3)$.
- **$\mathcal{X}$ -step (t-SVD):** one forward and one inverse FFT along mode-3, $\mathcal{O}(I_1 I_2 I_3 \log I_3)$; plus I_3 truncated SVDs of size $I_1 \times I_2$. With Lanczos/randomized SVD, each slice costs $\mathcal{O}((I_1 + I_2) I_{\min} R)$ where $I_{\min} = \min\{I_1, I_2\}$, yielding total $\mathcal{O}(I_3(I_1 + I_2) I_{\min} R)$. In square-ish cases this simplifies to $\mathcal{O}(I_1 I_2 R I_3)$.

Thanks to conjugate symmetry of the FFT for real tensors, only $\lfloor I_3/2 \rfloor + 1$ unique slices need full SVDs; the remaining slices can be mirrored, effectively halving the SVD cost. The memory footprint is $\mathcal{O}(I_1 I_2 I_3)$, and the SVDs across slices are embarrassingly parallel.

Algorithm 1 NTD-tSVD

Input: $\mathcal{M}$, $\mathcal{X}^0$, R , K .

Output: A rank- R tensor $\mathcal{X}$ s.t. $\mathcal{M} \approx \max(0, \mathcal{X})$.

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1: Set  $\mathcal{W}_{i_1 i_2 i_3}^k = \mathcal{M}_{i_1 i_2 i_3}$  for  $(i_1, i_2, i_3) \in \Omega_+$  for all  $k$ .
2: for  $k = 0, 1, \dots, K$  do
3:    $\mathcal{W}_{i_1 i_2 i_3}^{k+1} = \min(0, \mathcal{X}_{i_1 i_2 i_3}^k)$  for  $(i_1, i_2, i_3) \in \Omega_0$ .
4:    $\mathcal{D} = \text{fft}(\mathcal{W}^{k+1}, [], 3)$ .
5:   for  $i = 1, \dots, I_3$  do
6:      $[U, S, V] = \text{SVD}(\mathcal{D}(:, :, i))$ .
7:      $\hat{U}(:, :, i) = U$ ,  $\hat{V}(:, :, i) = V$ ,  $\hat{S}(:, :, i) = S$ .
8:   end for
9:    $\mathcal{U} = \text{ifft}(\hat{U}, [], 3)$ ,  $\mathcal{V} = \text{ifft}(\hat{V}, [], 3)$ ,  $\mathcal{S} = \text{ifft}(\hat{S}, [], 3)$ .
10:   $\mathcal{X}^{k+1} = \mathcal{U}(:, 1 : R, :) * \mathcal{S}(1 : R, 1 : R, :) * \mathcal{V}(:, 1 : R, :)$ 
11: end for
12:  $\mathcal{X} = \mathcal{X}^{k+1}$ .
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Convergence properties: Define

$$F(\mathcal{W}, \mathcal{X}) := \frac{1}{2} \|\mathcal{W} - \mathcal{X}\|_F^2 + \iota_{\mathcal{A}}(\mathcal{W}) + \iota_{\mathcal{B}_R}(\mathcal{X}),$$

where $\mathcal{A} := \{\mathcal{W} : \max(0, \mathcal{W}) = \mathcal{M}\}$. Algorithm 1 is a two-block alternating minimization scheme. For fixed $\mathcal{X}^k$, the $\mathcal{W}$ -update is the entrywise Euclidean projection of $\mathcal{X}^k$ onto the constraint $\mathcal{A}$, which yields the closed form in (5). For fixed $\mathcal{W}^{k+1}$, the $\mathcal{X}$ -update is exactly the truncated t-SVD step (best rank- R t-SVD approximation of $\mathcal{W}^{k+1}$). Thus both block updates are exact minimizers, and we have $F(\mathcal{W}^{k+1}, \mathcal{X}^k) \leq F(\mathcal{W}^k, \mathcal{X}^k)$, $F(\mathcal{W}^{k+1}, \mathcal{X}^{k+1}) \leq F(\mathcal{W}^{k+1}, \mathcal{X}^k)$, so $\{F(\mathcal{W}^k, \mathcal{X}^k)\}$ is nonincreasing and bounded below ($F^* \geq 0$). Assuming the initial level set is bounded, $\{(\mathcal{W}^k, \mathcal{X}^k)\}$ is bounded and admits cluster points. Moreover, F is proper, lower semicontinuous and semi-algebraic [25], and NTD-tSVD is a two-block scheme with exact updates. Hence standard nonconvex block coordinate descent theory (e.g., [25], [26], [27]) implies that every cluster point of $\{(\mathcal{W}^k, \mathcal{X}^k)\}$ is a stationary point of the constrained problem. Global optimality cannot be guaranteed due to non-convexity.

III. NUMERICAL EXPERIMENTS

We apply the proposed NTD-tSVD³ to address the model (4) and compare its performance with the tSVD⁴ [18], CP-ALS-QR [28], and Tucker-ALS [1], CP-ALS [1] and its non-negative version (CP-ALS-Prox), and CP-ALS for model (4) (denote CP-ALS-ReLU). We let $K = 100$ all algorithms. All algorithms use the same random initial point. The experiment is run 5 times under different initial points and reports the average result. All experiments are conducted in MATLAB on a PC running Ubuntu 24.04, equipped with an Intel CORE i7-14700KF @ 3.40GHz and 64GB RAM.

A. Synthetic dataset

To evaluate the effectiveness of the models and algorithms on nonnegative tensor data with varying sparsity, we generate sparse tensors $\mathcal{M}$ using the pseudo-code, i.e., $A_{(i)} \leftarrow$

$\text{randn}(I_i, \bar{R})$, $i = 1, 2, 3$, $\hat{\mathcal{M}} \leftarrow \sum_{r=1}^{\bar{R}} A_{(1)}(:, r) \circ A_{(2)}(:, r) \circ A_{(3)}(:, r)$, $\mathcal{M} \leftarrow \max(0, \hat{\mathcal{M}} - p\mathcal{M}_{\max})$, where $\mathcal{M}_{\max}$ is the maximum value of $\hat{\mathcal{M}}$, and $p \in [0, 1]$ controls the sparsity of $\mathcal{M}$. If $p = 0$, then the sparsity is 50% (the corresponding proportion (%) of zero elements). If $p = 1$, the corresponding sparsity is 100%.

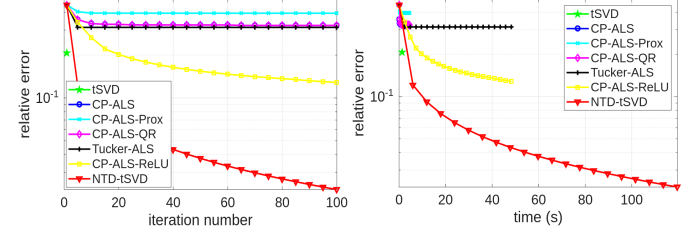


Fig. 3: Numeric results for the synthetic dataset with $I_1 = 200$, $I_2 = 200$, $I_3 = 250$, $\bar{R} = 11$ and $p = 0.4$ under $R = 25$.

Table I shows that NTD-tSVD consistently and substantially outperforms the baselines across all tensor sizes, ranks, and sparsity levels. Across all settings, increasing the target t-rank R monotonically improves reconstruction for all methods. From Table I, tSVD-based methods have higher per-iteration cost, but Fig. 3 shows that, for the same wall-clock time, NTD-tSVD attains the lowest reconstruction error.

This aligns with the modeling intuition: when many entries of $\mathcal{M}$ are zero due to rectification, explicitly disentangling the ReLU nonlinearity (via the $\mathcal{W}$ -projection) and the low-rank prior (via truncated tSVD) is crucial, whereas direct low-rank approximation of $\mathcal{M}$ alone is structurally mismatched; the proposed splitting prevents “smearing” of sparse patterns and reduces bias on rectified zeros (see Fig. 3).

B. Real dataset

In this subsection, we evaluate the performance and stability of the proposed model on five (sparse) nonnegative real-world tensors from neuroscience, chemometrics, computer vision, and communication. The datasets are:

- *BMI*⁵: a Monkey Brain-Machine Interface spike-count tensor of size $43 \times 200 \times 88$ with sparsity 13.98%.
- *EEM*⁶: an excitation-emission fluorescence tensor of size $18 \times 251 \times 21$ with sparsity 23.11%.
- *COIL20*⁷: a grayscale image tensor built from 20 object database, of size $1024 \times 72 \times 20$ with sparsity 34.39%.
- *MNIST*⁸: a handwritten digit image tensor of size $784 \times 5000 \times 10$ with sparsity 80.73%.
- *Emails*⁹: an Enron email count tensor of size $105 \times 105 \times 28$ with sparsity 97.23%.

Table II summarizes results on five real datasets, and Fig. 4 visualizes representative slices for *MNIST*, showing that NTD-tSVD preserves salient patterns while suppressing spurious activations. Figures 5 and 6 only present the decomposition

⁵https://gitlab.com/tensors/tensor_data_monkey_bmi

⁶https://gitlab.com/tensors/tensor_data_eem

⁷<https://www.cs.columbia.edu/CAVE/software/softlib/coil-20.php>

⁸<http://yann.lecun.com/exdb/mnist/>

⁹https://gitlab.com/tensors/tensor_data_enron_emails

TABLE I: Relative errors on synthetic data for different tensor sizes I_i , ranks $\bar{R}$, and sparsity levels. The first row gives $\dim(\mathcal{M})$, the second the sampling rate p and the corresponding percentage of zero entries, and the third the rank R .

Algorithms	$50 \times 80 \times 100, R = 10$				$200 \times 200 \times 250, R = 11$				$350 \times 350 \times 350, R = 12$				Average
	$p = 0.1, 58.81\%$		$p = 0.3, 74.98\%$		$p = 0.4, 85.98\%$		$p = 0.5, 91.11\%$		$p = 0.6, 95.56\%$		$p = 0.8, 98.78\%$		time (s)
	12	15	20	30	25	40	40	60	60	80	80	100	per iteration
tSVD	1.44e-1	1.18e-1	1.10e-1	5.91e-2	2.07e-1	1.54e-1	1.67e-1	1.16e-1	1.99e-1	1.61e-1	1.61e-1	1.26e-1	2.33
CP-ALS	2.75e-1	2.47e-1	2.58e-1	2.30e-1	3.24e-1	2.94e-1	3.24e-1	2.94e-1	3.65e-1	3.49e-1	3.99e-1	3.87e-1	0.07
CP-ALS-Prox	3.66e-1	3.48e-1	3.54e-1	3.21e-1	3.94e-1	3.58e-1	3.73e-1	3.66e-1	4.07e-1	4.07e-1	4.34e-1	4.31e-1	0.14
CP-ALS-QR	2.75e-1	2.47e-1	2.58e-1	2.30e-1	3.24e-1	2.94e-1	3.24e-1	2.94e-1	3.65e-1	3.49e-1	3.99e-1	3.87e-1	0.14
Tucker-ALS	2.49e-1	2.15e-1	2.52e-1	2.06e-1	3.13e-1	2.81e-1	3.10e-1	2.69e-1	3.47e-1	3.21e-1	3.73e-1	3.50e-1	0.15
CP-ALS-ReLU	2.83e-2	2.50e-2	6.64e-2	5.81e-2	1.29e-1	1.09e-1	1.46e-1	1.35e-1	2.03e-1	1.91e-1	2.71e-1	2.54e-1	1.95
NTD-tSVD	1.89e-2	1.40e-2	1.17e-3	4.78e-6	2.27e-2	3.42e-3	1.34e-3	4.89e-5	1.21e-3	2.05e-4	1.37e-4	3.29e-5	2.36

TABLE II: The relative errors under different R for five real datasets. The second row indicates the value of R .

Algorithms	<i>BMI</i>			<i>EEM18</i>			<i>COIL20</i>			<i>MNIST</i>			<i>Emails</i>			Average time (s) per iteration
	10	20	30	2	3	5	20	30	40	30	40	60	15	20	28	
tSVD	1.26e-1	1.37e-3	7.16e-4	4.59e-2	8.91e-3	7.10e-3	4.64e-2	3.22e-2	2.27e-2	1.99e-1	1.77e-1	1.47e-1	1.66e-1	1.37e-1	1.00e-1	3.22
CP-ALS	1.71e-1	1.52e-1	1.39e-1	6.95e-2	2.64e-2	1.43e-2	1.40e-1	1.25e-1	1.14e-1	2.86e-1	2.73e-1	2.53e-1	2.46e-1	2.24e-1	1.99e-1	0.16
CP-ALS-Prox	1.80e-1	1.65e-1	1.58e-1	7.06e-2	1.66e-2	1.13e-2	1.79e-1	1.72e-1	1.65e-1	2.94e-1	2.85e-1	2.76e-1	2.51e-1	2.31e-1	2.27e-1	0.18
CP-ALS-QR	1.71e-1	1.52e-1	1.39e-1	6.95e-2	2.64e-2	1.43e-2	1.40e-1	1.25e-1	1.14e-1	2.86e-1	2.73e-1	2.53e-1	2.46e-1	2.24e-1	1.99e-1	0.25
Tucker-ALS	1.63e-1	1.38e-1	1.16e-1	6.95e-2	1.63e-2	1.05e-2	1.17e-1	1.01e-1	9.01e-2	2.82e-1	2.70e-1	2.50e-1	2.34e-1	2.06e-1	1.70e-1	0.28
CP-ALS-ReLU	1.70e-1	1.52e-1	1.38e-1	6.89e-2	2.52e-2	1.41e-2	1.31e-1	1.16e-1	1.05e-1	2.79e-1	2.63e-1	2.41e-1	2.04e-1	1.80e-1	1.52e-1	0.72
NTD-tSVD	1.06e-2	1.19e-3	4.96e-4	4.21e-2	8.56e-3	6.37e-3	3.80e-2	2.54e-2	1.72e-2	1.29e-1	1.06e-1	7.96e-2	1.06e-2	4.30e-3	1.05e-3	3.28

results for several images from the two datasets obtained using the two algorithms, since these two algorithms exhibit better numerical performance in Table II. A similar accuracy–runtime trade-off is observed on real data. As shown in Table II, the tSVD-based methods are more expensive per iteration, whereas Fig. 4 demonstrates that NTD-tSVD attains the best reconstruction quality within the same wall-clock time. In practice, we can use a maximum runtime as the stopping criterion for the algorithm.

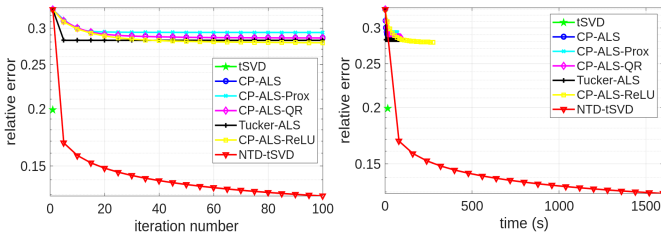


Fig. 4: Numerical results for the *MNIST* dataset under $R = 30$.

For example, Fig. 6 directly contrasts ReLU-based reconstructions at $R=60$: compared with tSVD, NTD-tSVD reaches better PSNR value, yielding visibly crisper digit strokes and cleaner backgrounds—evidence that recovering the latent pre-rectified structure translates into better fidelity after rectification. Beyond *MNIST* dataset, the advantage is pronounced on highly sparse *Emails* dataset, and similar improvements appear on *BMI*, *EEM18*, and *COIL20* datasets at comparable ranks (see Table II details).

Across synthetic and real data, NTD-tSVD (Algorithm 1) yields up to orders-of-magnitude lower relative error than baselines as sparsity increases. These results indicate that explicitly separating the ReLU nonlinearity from the low-rank prior delivers both quantitative and perceptual benefits under practical sparsity levels.

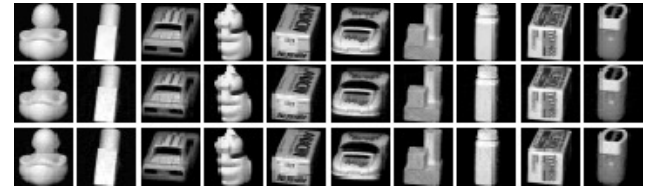


Fig. 5: Low-rank approximation for the *COIL20* dataset with $\text{rank}(\mathcal{X}) = 40$. **Top**: original data. **Middle**: model (1) via tSVD [17], [18], PSNR= 34.03. **Bottom**: model (2) via Algorithm 1, PSNR= 36.79.



Fig. 6: Low-rank approximation for the *MNIST* dataset ($784 \times 5000 \times 10$) with $\text{rank}(\mathcal{X}) = 60$. **Top**: original data. **Middle**: model (1) via t-SVD [17], [18], PSNR= 20.10. **Bottom**: model (2) via Algorithm 1, PSNR= 26.30.

IV. CONCLUSION AND FUTURE WORK

We proposed a ReLU-based nonlinear tensor decomposition model and an alternating algorithm (Algorithm 1) that separates the element-wise nonlinearity from the t-rank constraint. Computational complexity and convergence are also analyzed. Extensive experiments on synthetic and real tensors demonstrate consistent gains over tSVD baselines—especially in highly sparse regimes. Several problems can be considered for future work, such as other nonlinear function [29], criteria and efficient heuristics for estimating rank R [30], and ReLU-based matrix/tensor completion [14].

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